

Spur-Free MASH Delta-Sigma Modulation

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Abstract—For multistage noise-shaping (MASH) delta-sigma modulation, this paper presents a new structure that is free of spurs for all input values. The proposed MASH structure cascades several first-order delta-sigma modulators (DSMs) like the traditional MASH structure but has an additional feedforward connection between two adjacent stages. The proposed MASH structure can increase the sequence length and thus reduce spurs. The reason why the proposed MASH structure has a long sequence length for the full input range is mathematically proved, and simulations are performed to verify the effect of the long sequence length. Simulation results show that the performance of the proposed MASH structure is close to that of the ideal DSM. In addition, the proposed MASH structure requires almost the same hardware cost as the traditional MASH structure.

Index Terms—Delta-sigma modulator (DSM), fractional- N frequency synthesizer, Multistage noise shaping (MASH).

I. INTRODUCTION

DELTA-SIGMA modulation has found increasing use in a wide range of modern electronic components, such as frequency synthesizers, analog-to-digital and digital-to-analog converters, and switched-mode power supplies. Delta-sigma modulation translates a fine input sequence that should be represented in m bits to a coarse output sequence represented in n bits, where n is significantly less than m . The delta-sigma modulator (DSM) is an oversampling converter in the sense that the sampling frequency is higher than twice the Nyquist frequency of the input sequence. As a result, the quantization error power is spread over wider frequencies and filtered such that the noise power is concentrated on high frequencies. The DSM usually generates a binary sequence. Let the input be a low-frequency tone of f_0 represented in b bits, as shown in Fig. 1(a). The spectrum of the output sequence is shown in Fig. 1(b). Note that the spectrum of the binary output sequence contains not only the full information of the input sequence but also the quantization error noise denoted by the crosshatched area. Ideally, the resulting quantization error should be whitened and filtered so that its power is moved away from the signal through a process called noise shaping. As shown in Fig. 1(b), the shaped noise is away from f_0 and concentrated around $f_s/2$, where f_s is the sampling frequency of the DSM. By passing the output signal through a low-pass filter, the original signal can be recovered. In the spectrum of the low-pass filter output shown in Fig. 1(c), the signal power is much higher than the

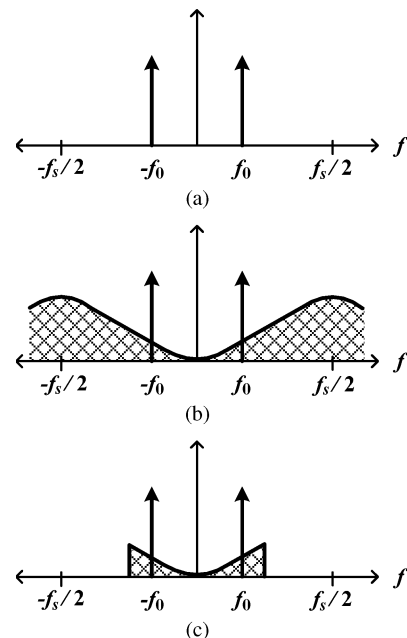


Fig. 1. Delta-sigma modulation of a sinusoidal signal. (a) Power spectrum of the sinusoidal tone. (b) Power spectrum of the output sequence. (c) Low-pass-filtered spectrum.

noise power. The degree to which the quantization noise can be attenuated by a low-pass filter depends on the order of the noise shaping and the oversampling ratio.

The DSM is widely used to select a clock divider in the fractional- N frequency synthesizer [1]–[3], [10]–[12]. Let us synthesize a frequency of $N \cdot \alpha$ times the reference frequency, where N is an integer and α is a fractional value. In that case, the DSM output sequence is used to select one of two dividers, with each of which divides the VCO frequency by $(N + 1)$ or N . If the $(N + 1)$ divider is selected when the DSM output is one, the average of the DSM output sequence should be α . The DSM input is set to a constant value that can produce an output average of α . For most frequency synthesizers, the DSM input is usually generated by a digital hardware unit designed based on the normal binary number system. In addition, the input value is a constant digital word represented in b bits so that the input value is ranged from 0 to $2^b - 1$. It is sometimes important to support the full input range of the DSM, particularly for frequency-hopping applications. The output of the DSM should be a sequence of which the average value is linearly proportional to the constant input. If this constraint is satisfied, we call the DSM accurate. If the output average has linearly related to the input value, additional computation or hardware circuitry is required to calculate the input value from the desired fractional value α . Moreover, the converted value may require a longer bitwidth than that of α . If the average value of the output sequence is

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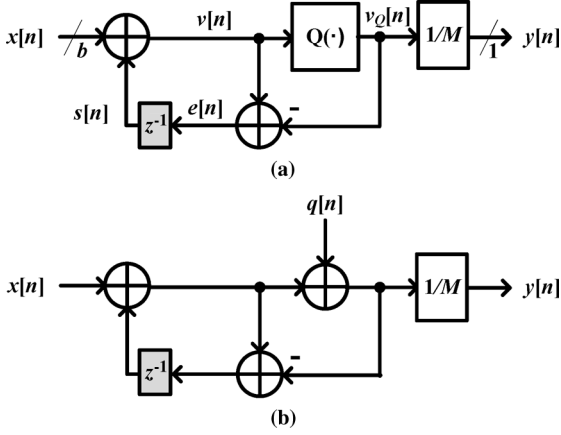


Fig. 2. (a) Block diagram of the EFM. (b) Mathematical model with an additive quantization noise source $q[n]$.

equal to the input value, the DSM does not require any conversion operation, and the bitwidth of the input value can be set to that of α .

As shown in Fig. 2, the first-order DSM consists of a digital accumulator, a register, and a quantizer. Since the DSM is a finite-state machine (FSM), its output sequence is periodic when the input is a constant value. More importantly, the quantization noise sequence is also periodic, and the noise sequence period or the sequence length is strongly dependent on the input value and the DSM structure. If the sequence length is short, the quantization noise power is spread over a small number of tones, resulting in spurs in the noise spectrum [4]. The purity of the output frequency is directly affected by the spectrum imperfection, especially by the spurs. To enhance the noise-shaping capability in the in-band region, several DSMs can be cascaded. The multistage noise shaping (MASH) structure is commonly employed because it is inherently stable [3]. However, the output spectrum of the traditional MASH structure still has spurs since the sequence length is very short for some input values.

One way to increase the sequence length is to apply a dither signal to the input value or the accumulator output. Dithering breaks up the periodicity and increases the effective sequence length, resulting in a smoothly noise-shaped spectrum. Although it is effective in increasing the sequence length as desired, the dithering inherently adds noise to the spectrum and makes the DSM unstable [9]. To resolve such problems, deterministic methods have been presented [4],[6],[7]. Setting the initial value of the modulator register to an odd value was proposed in [4], but the sequence length rarely improves even if the MASH order increases. In [6], the quantization interval is changed to a prime number in order to increase the minimum sequence length of the first-order DSM to the prime number. For the MASH structure, however, the sequence length is only defined by the first stage, which means that the following stages do not contribute to the sequence length. Recently, a modified DSM that has an additional feedback loop has been presented in [7], which increases the sequence length exponentially with respect to the MASH order. When adopting b -bit DSM, the input range cannot cover all b -bit input values. Although the modified DSM is accurate, the average value of the output

sequence is not equal to the input value because the average of the output sequence deviates from the input value.

There are three factors to be considered in implementing a DSM. The first factor is related to the average value of the output sequence and the input value. If the two values are linearly related to each other, the DSM is accurate. The second factor is the sequence length, which should be long enough to smoothly distribute the power of the quantization error. The last is the input range. The DSM should support the full input range since most applications are required. The ideal DSM would be accurate and would generate the infinite sequence lengths for all input values.

This paper proposes a new MASH structure that is very close to the ideal DSM in terms of accuracy, input range, and sequence length. Moreover, the proposed MASH structure does not increase the hardware cost even compared to the traditional MASH structure. The rest of this paper is organized as follows. Section II describes the overall operation of the DSM and summarizes previous works. Section III presents the proposed MASH structure, and Section IV describes the simulation results and comparisons with other works. Concluding remarks are made in Section V. Finally, mathematical proofs are provided in the Appendix.

II. PREVIOUS DSMs

A typical structure of the first-order DSM, which corresponds to the first-order error feedback modulator (EFM), includes a quantizer, a register, and an accumulator with a feedback loop, as shown in Fig. 2(a). The first-order EFM is widely used in implementing the first-order DSM, since the EFM has a relatively low hardware cost and low power consumption [19]. For the sake of simplicity, the first-order EFM is referred to as EFM throughout this paper. In most applications, the input of the DSM is a fractional value in the range of $[0, 1)$ [14]. The input fractional value usually generated by a controller is a fixed-point binary number represented in b bits, and the radix point is assumed to be located at the left of the most significant bit. For implementing the DSM, the input is shifted left by b bits, which means that the radix point is moved to the right of the least significant bit. Therefore, the input $x[n]$ is a b -bit integer value ranging from 0 to $M - 1$, where $M = 2^b$. The quantizer $Q(\cdot)$ compares the value of the digital accumulator with a threshold value T . If $v[n]$, which is the sum of $x[n]$ and $s[n]$, is equal to or greater than T , the quantizer output $v_Q[n]$ will be M ; otherwise, it is zero as follows:

$$v_Q[n] = Q(v[n]) = \begin{cases} 0, & v[n] < T \\ M, & v[n] \geq T. \end{cases} \quad (1)$$

Since the quantization interval is M , the output $y[n]$ is derived by dividing $v_Q[n]$ by M or shifting $v_Q[n]$ right by b bits. The difference between $v[n]$ and $v_Q[n]$ is sampled by a gray-colored register to feed it back to the input at the next cycle. If $y[n]$ is zero, then $e[n] = v[n]$; otherwise, $e[n] = v[n] - M$. Although the threshold value T can be any value equal to or less than M , the threshold value T is usually set to M for the sake of simple implementation. In this case, $e[n]$ can be expressed as

$$e[n] = v[n] \bmod M. \quad (2)$$

In addition, the carryout signal of the digital accumulator becomes $y[n]$. When the quantizer is replaced with an additive quantization noise source $q[n]$ as shown in Fig. 2(b), the output signal $y[n]$ is expressed as

$$y[n] = \frac{1}{M} (x[n] + q[n] - q[n-1]). \quad (3)$$

If the period of the EFM is N

$$\sum_{n=1}^N y[n] = \frac{1}{M} \sum_{n=1}^N x[n] + (q[N] - q[0]). \quad (4)$$

Since $q[0] = q[N]$, the average of $y[n]$ [$Mean(Y)$] is equal to the average of $x[n]$ [$Mean(X)$] divided by the modulus M

$$Mean(Y) = \frac{1}{M} Mean(X). \quad (5)$$

According to (5), the average of $y[n]$ is dependent on the average of the input digital word $x[n]$ and the modulus M . The average is also equal to the desired fractional value so that the EFM is accordingly accurate.

The input of the EFM is a b -bit digital word which is the same as the input fractional value, but the radix point is moved to the right of the least significant bit. Since $Mean(X)/M$ is equal to the normal binary number of which radix point is located at the left of the most significant bit, the mean of $y[n]$ is straightforwardly equal to the input fractional value. Moreover, the EFM supports the full input range from 0 to $M - 1$.

We can write the z -domain output $Y(z)$ in terms of $X(z)$ and $Q(z)$ as follows:

$$Y(z) = \frac{1}{M} X(z) + \frac{1}{M} (1 - z^{-1}) Q(z). \quad (6)$$

Therefore, the EFM includes two filters, an all-pass filter for the input $X(z)$ and a high-pass filter for the quantization error $Q(z)$. The filtering functions related to the input signal and the quantization error are called signal transfer function (STF) and noise transfer function (NTF), respectively. In (6), the STF is $1/M$, and the NTF is $(1 - z^{-1})/M$. Note that the output spectrum of the EFM includes the full information of the input sequence. If we assume that $q[n]$ has a white noise spectrum, the spectrum of the output quantization error is a smooth curve shaped by the filter $(1 - z^{-1})/M$.

Since the EFM is an FSM, $y[n]$ and $e[n]$ are periodic if $x[n]$ is periodic. For a constant input value, the output sequence length of the EFM is determined by Theorem 1 [13].

Theorem 1: The sequence length N of the EFM is $N = 2^{b-j} = M/2^j$ if the input signal $x[n]$ is $M \cdot \alpha$, and $M \cdot \alpha$ is a rational number, $M \cdot \alpha = \sum_{k=0}^{b-1} p_k \cdot 2^k = 2^j \cdot B$, where $B \bmod 2 = 1$, p_k is zero or one, and $2^j < M$.

The proof is described in the Appendix. Note that the sequence length depends on the input value. For some inputs, therefore, the sequence length is relatively short, and the noise power is concentrated on a few tones. The output spectrum accordingly contains spurs, deviating significantly from the smoothly noise-shaped curve.

In order to achieve a better NTF, the MASH structure shown in Fig. 3 has been developed by cascading a number of EFMs.

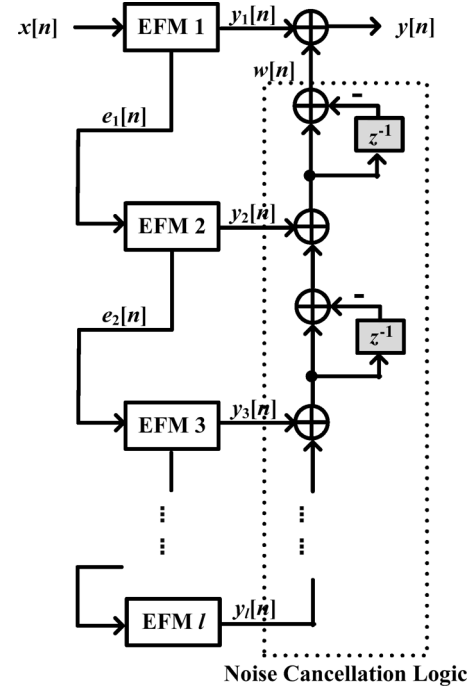


Fig. 3. Traditional MASH DSM with l stages.

The error of each stage $e_i[n]$ is connected to the next stage, and the carryout of each stage $y_i[n]$ is fed toward the noise cancellation logic to eliminate the intermediate quantization noise. The spectrum of the final output $y[n]$ includes the spectrum of the input signal $x[n]$ as well as the shaped quantization noise power of the last stage. In the z -domain, the linear model of a traditional MASH structure [8] can be expressed as

$$Y(z) = \frac{1}{M} X(z) + \frac{1}{M} (1 - z^{-1})^l Q_l(z) \quad (7)$$

where Q_l represents the quantization error of the last stage. In (7), we can see that the input signal $X(z)$ appears directly in the output spectrum, and the quantization noise of the last stage $Q_l(z)$ is shaped by the filter $(1 - z^{-1})^l$, where l is the MASH order representing the number of stages.

The average of the noise cancellation logic output $w[n]$ can be derived. Suppose that $b[n] = a[n] - a[n-1]$. In this case, the average of $b[n]$ is zero if $a[n]$ is a periodic signal. As the $y_2[n]$ and $y_3[n]$ that are shown in Fig. 3 are periodic signals, $w[n]$ is a periodic signal whose average is zero. Thus, the average of the MASH output is equal to the mean of the output of the first-stage EFM. Since $y_1[n]$ and $w[n]$ are periodic, the sequence length of $y[n]$ is the least common multiple (LCM) of the sequence lengths of $y_1[n]$ and $w[n]$.

The MASH structure can increase the order of noise shaping, as shown in (7), but it suffers from spurs resulting from non-white short sequences. The sequence length should be long enough to achieve smooth spectra of the output $Y(z)$ for all possible input values. However, the sequence length resulting from the MASH structure is sometimes very short depending on the input value.

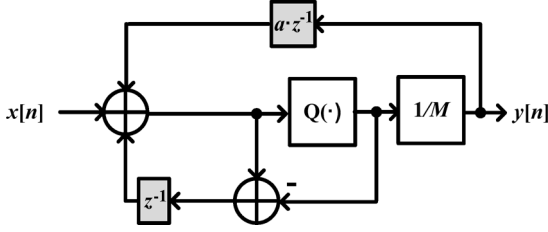


Fig. 4. HK-EFM structure.

As mentioned previously, the ideal DSM is desired to be accurate, and it supports infinite sequence lengths for all input values. For the traditional MASH DSM, the average of the output sequence is exactly equal to the input value, but the output sequence length is too short for some input values. To resolve the problem, the stochastic approach [3],[5] has been presented to maximize the sequence length by adding a random dither sequence generated by a pseudorandom or chaotic source. The random sequence disrupts the sequence length to increase the period, resulting in smooth noise-shaped spectra. However, the approach inherently adds noise to the in-band spectrum [5]. The in-band noise adversely affects the phase noise performance of the frequency synthesizer as the low-pass filter cannot remove the in-band noise [10]. Furthermore, the stochastic approach affects the DSM stability and needs extra hardware resources to implement the random sequence [9]. Another way to whiten the quantization noise is the deterministic approach that guarantees long sequence lengths by design. A simple method is to restrict the initial value of a register to an odd value [4]. Although this method support relatively long sequence lengths compared to the traditional MASH DSM, the sequence length rarely improves even when the MASH order increases. Employing a prime number P for the modulus is another method [6]. This method guarantees that the sequence length is equal to P , but the input range is reduced to $\{0, 1, \dots, (P - 1)\}$. The state-of-the-art deterministic method, denoted as the HK-EFM [7], is shown in Fig. 4. In this method, an additional backward connection is used to feed the output value multiplied by a to the input side. The value of a is determined depending on the input bitwidth, and it effectively makes the modulus be a prime number $(M - a)$. Accordingly, most input values have a sequence length of $(M - a)$. If the MASH order is l , the sequence length is increased to $(M - a)^l$. Therefore, the MASH structure based on the HK-EFM, denoted as the HK-MASH, guarantees the long sequence length. However, the input range of the HK-EFM is bounded up to $(M - a - 1)$ as the effective quantization interval is $(M - a)$. In addition, the additional backward connection disrupts the input value. Due to the mean of the feedback connection $Mean(Y) \cdot a$, the actual output average of the HK-EFM converges to $M \cdot \alpha / (M - a)$ if the input fractional value is α . More specifically, the HK-EFM satisfies $Mean(Y) = Mean(X) / (M - a)$. For the HK-MASH, Fig. 5 shows how the average of the output sequence is different from the input fractional value. Since the difference is proportional to the input fractional value, the output average is linearly related to the input value. If we want to make the output average equal to α , the input value should be scaled down to $\alpha(M - a)/M$.

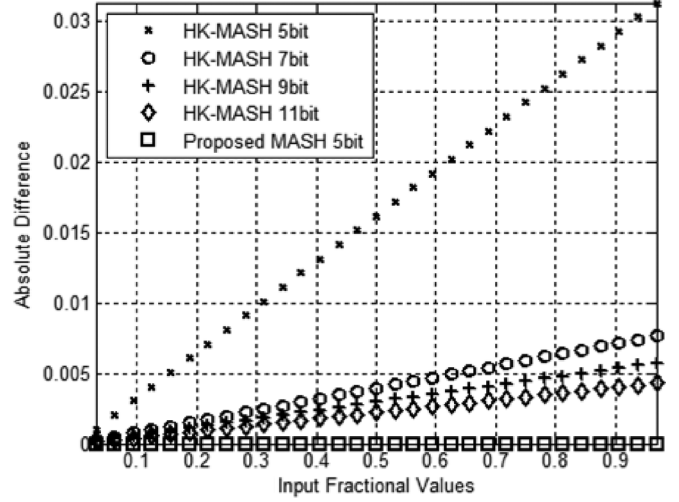


Fig. 5. Absolute difference between the input fractional values and the output averages of the HK-MASH and the proposed MASH structures.

This is the reason why the HK-MASH does not support the full input range. Additional computation or hardware circuitry is required for the scaling, and the scaled value usually requires a longer bitwidth than that of α . We propose a new MASH structure to make the average of the output sequence exactly equal to the input fractional value, as shown in Fig. 5.

III. PROPOSED MASH DSM

The stochastic and deterministic methods are inherently adding a nonzero mean signal to the input. As the nonzero mean signal makes the output average deviate from the input value, the average of the output sequence is not equal to the input value. Thus, the simplest method to match the input value and the output average is not to add any nonzero mean signal to the input. In this section, we propose a new MASH structure that totally stands in matching the input value and the output average and that provides long sequence lengths for the full input range. As shown in Fig. 6, the proposed MASH structure is different from the traditional MASH structure. The traditional MASH structure cascades a number of EFMs by connecting the quantization error signal of an EFM to the input of the next-stage EFM, whereas the proposed MASH structure links the EFMs by connecting the output and the quantization error signal of an EFM to the input of the next-stage EFM. In the proposed MASH structure, hence, the output of an EFM is delivered to not only the noise cancellation logic but also the subsequent stage. Each EFM constituting the proposed MASH structure takes two inputs and generates two outputs to be fed to the next EFM, as shown in Fig. 7.

It is important to note that the input of the first stage is a pure dc signal and not a perturbed one. Any input value is allowed within the quantization interval, and thus, the proposed MASH supports the full input range $\{0, 1, \dots, (M - 1)\}$. In the proposed EFM, the output and the quantization error signals that are to be connected to the next stage have the same sequence length. Since two periodic inputs can be regarded as one periodic signal, the proposed EFM is functionally equivalent to the traditional EFM, and it generates a periodic output signal. In the

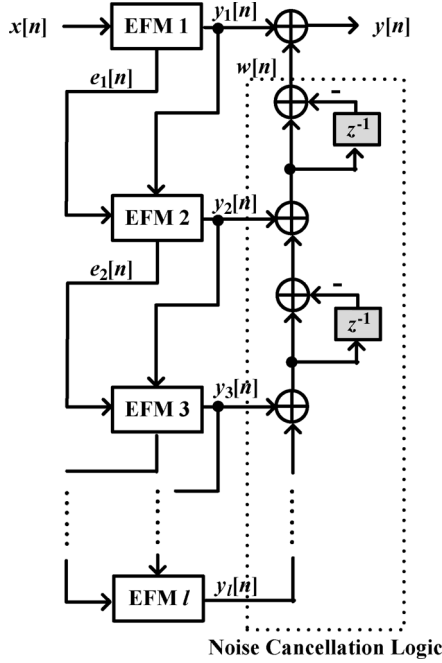
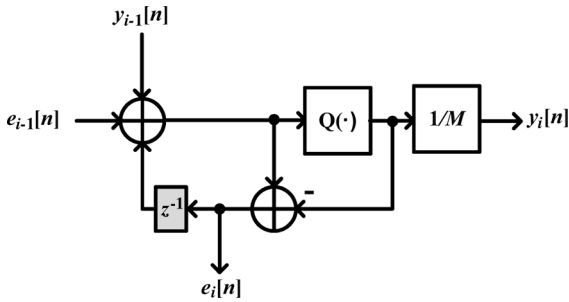


Fig. 6. Proposed MASH structure.

Fig. 7. Proposed EFM structure of stage i .

noise cancellation logic, $w[n]$ is periodic, and its average is always zero. Hence, the average of $y[n]$ in the proposed MASH is accordingly equal to the mean of the first-stage output.

Now, let us derive the sequence length of the proposed MASH DSM. We start from the third-order MASH denoted as MASH 1-1-1. In Theorem 1, the sequence length N_1 of the first stage is explicitly related to the constant input. The sequence lengths of the second and third stages can be derived by Proposition 1, which will be proved in the Appendix.

Proposition 1: The sequence lengths of the second and the third stages of the proposed MASH 1-1-1 structure are $N_1 \cdot M$ and $N_1 \cdot M^2$, respectively, if the input signal is constant.

The sequence length of MASH 1-1-1 is the LCM of the sequence lengths of the three stages, i.e., $N_1 \cdot M^2$. Moreover, Proposition 1 is satisfied irrespective of the initial value of the register denoted as z^{-1} in Fig. 7. In general, the sequence length of the l th-order MASH structure can be determined by the following theorem.

Theorem 2: The sequence length of the proposed l th-order MASH structure is $N_1 \cdot M^{l-1}$ if the input signal is constant.

The sequence length of the proposed MASH structure increases exponentially with respect to the MASH order. Since Theorem 2 is satisfied irrespective of the initial values of the registers, the input value can be changed in the middle of normal operation without resetting the registers beforehand. The proposed MASH structure is very close to the ideal DSM as it is accurate and supports long sequence lengths for all input values.

The STF and the NTF of the proposed MASH structure should be analyzed. The STF and the NTF for a single EFM are $1/M$ and $(1 - z^{-1})/M$, respectively. Using the results of the single EFM, the z -transform of the output at each stage is

$$Y_1(z) = \frac{1}{M}X(z) + \frac{1}{M}(1 - z^{-1})Q_1(z) \quad (8)$$

$$Y_2(z) = \frac{1}{M}(Y_1(z) - Q_1(z)) + \frac{1}{M}(1 - z^{-1})Q_2(z). \quad (9)$$

For the l th stage, we obtain

$$Y_l(z) = \frac{1}{M}(Y_{l-1}(z) - Q_{l-1}(z)) + \frac{1}{M}(1 - z^{-1})Q_l(z). \quad (10)$$

Using the aforementioned equations and the structure shown in Fig. 6, the output of the proposed MASH is

$$Y(z) = Y_1(z) + (1 - z^{-1})Y_2(z) + \dots + (1 - z^{-1})^{l-1}Y_l(z) \quad (11)$$

$$Y(z) = \left\{ \frac{1}{M} + \frac{1}{M^2}(1 - z^{-1}) + \dots + \frac{1}{M^l}(1 - z^{-1})^{l-1} \right\} X(z) + (1 - z^{-1})^l \left\{ \frac{1}{M}Q_l(z) + \frac{1}{M^2}Q_{l-1}(z) + \dots + \frac{1}{M^l}Q_1(z) \right\}. \quad (12)$$

Finally, we have

$$Y(z) = \frac{1}{M}X'(z) + \frac{1}{M}(1 - z^{-1})^l Q'(z) \quad (13)$$

where

$$X'(z) = \left\{ 1 + \frac{1}{M}(1 - z^{-1}) + \dots + \frac{1}{M^{l-1}}(1 - z^{-1})^{l-1} \right\} X(z) \quad (14)$$

$$Q'(z) = Q_l(z) + \frac{1}{M}Q_{l-1}(z) + \dots + \frac{1}{M^{l-1}}Q_1(z).$$

Therefore, the STF and NTF of the proposed MASH structure with l stages are

$$STF(z) = \frac{1}{M} + \sum_{n=2}^l \frac{1}{M^n}(1 - z^{-1})^{n-1} \quad (15)$$

$$NTF(z) = \frac{1}{M}(1 - z^{-1})^l. \quad (16)$$

The STF expressed in (15) includes an all-pass filter and several high-pass filters. As the input signal is constant, the terms related to the high-pass filters never affect the performance of the modulator, which means that the STF can be regarded as an all-pass filter. Moreover, the quantization noise is shaped by the exact NTF expressed in (16).

Proposition 1 and Theorem 2 assume that the bitwidths are equal for all stages so that the minimum sequence length is

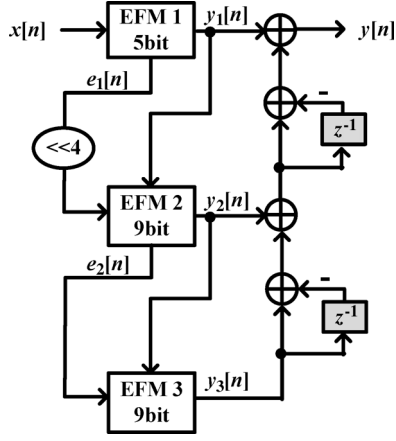


Fig. 8. Proposed MASH structure having one 5-bit EFM at the first stage and two 9-bit EFMs at the later stages.

$2M^2$. In order to achieve a longer sequence length, we can increase the bitwidths, except that of the first stage. In connecting $e_1[n]$ generated in the first stage to the second stage, it is left justified by the bitwidth difference of the first and second stages. As a result, the minimum sequence length of the proposed MASH is increased to $2L^2$, where $L = 2^r$, and r is the bitwidth of the later stages. This is described in Corollary 1.

Corollary 1: If the subsequent stages have a longer bitwidth r than that of the first stage b and the quantization error of the first stage $e_1[n]$ to be connected to the second stage is shifted left by the bitwidth difference $(r - b)$, the sequence length of the proposed l th-order MASH structure is $N_1 \cdot L^{l-1}$, where $L = 2^r$.

Fig. 8 shows an example MASH DSM of Corollary 1. When the input value is 5 bits, the modulus of the first stage is 32, and the modulus of the later stages is 512. In this configuration, the minimum sequence length is increased to 2^{19} .

The analysis of the STF and the NTF corresponding to Corollary 1 is very similar to the previous analysis because we can assume that the first stage has the same bitwidth as those of the later stages. In that case, the bitwidth of the first stage is enlarged to r , but a multiple of 2^{r-b} is only allowed as an input value. The resulting STF and NTF are exactly the same as (15) and (16) except that M is replaced with L .

IV. SIMULATION RESULTS AND COMPARISON

The period of a periodic signal can be predicted by referring to the peak index in the autocorrelation of the signal. Autocorrelation is a common technique that is used to detect the period of a periodic signal in many applications [16],[17]. To investigate the sequence length of the proposed MASH structure, we perform autocorrelation for the output signal of each stage, as shown in Fig. 9, which shows the autocorrelation results for the proposed MASH 1-1-1 structure having three 5-bit EFMs. The sequence length of the output signal is determined by the index whose peak value is equal to the peak value at index 0.

The sequence length of the first stage is dependent on the input value, and the minimum sequence length occurs when the input value is 16. In that case, the sequence length of the first stage is two, as shown in Fig. 9(a). The sequence lengths of the

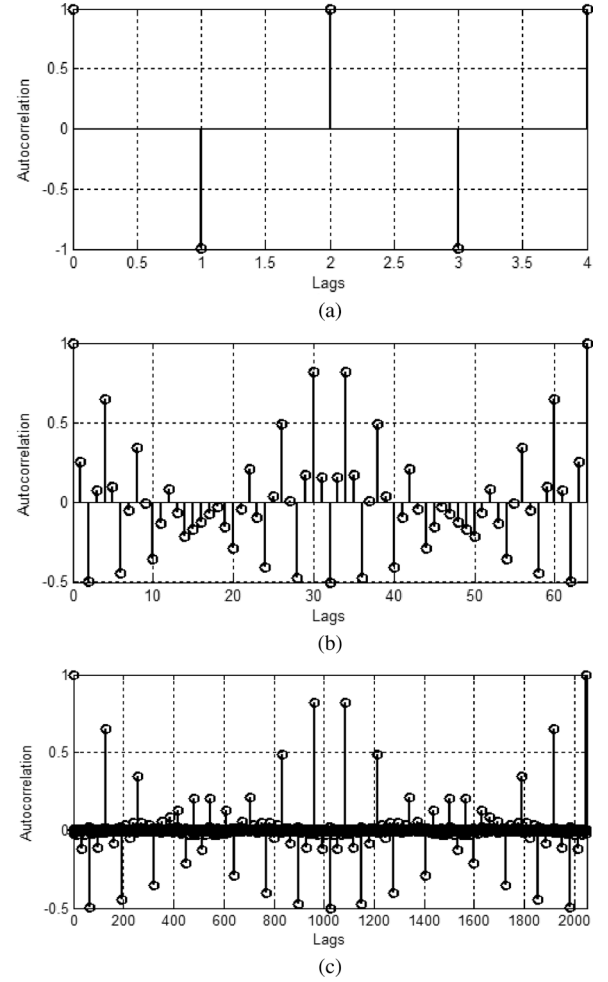


Fig. 9. Autocorrelations of the (a) first, (b) second, and (c) third stage outputs in the proposed MASH 1-1-1 structure having three 5-bit EFMs when the input value is 16.

second and third stages are also shown in Fig. 9(b) and (c). Note that the horizontal scales are different. The autocorrelation of the second-stage output has a peak at 64, and that of the third-stage output is at 2048. Adding a stage increases the sequence length by M , as is indicated by Proposition 1.

The longer sequence length leads to the wider distribution of the quantization noise power. For input values of 16 and 17 represented in 5 bits, the spectra of the HK-MASH 1-1-1 and the proposed MASH 1-1-1 are calculated with 2^{20} output samples, as shown in Fig. 10. The spectra of the HK-MASH 1-1-1 structure with 5-bit EFMs are shaped smoothly by the NTF, as shown in Fig. 10(a) and (c). The dashed line represents the NTF curve, and the solid line denotes the output spectra of the HK-MASH. The HK-MASH structure effectively removes spurs while preserving the noise-shaping property. Fig. 10(b) and (d) are the spectra resulting from the proposed MASH 1-1-1 with 5-bit EFMs. For several input values, Fig. 11 shows the output spectra of the proposed MASH with three 5-bit EFMs. The spectra are smooth, and there are no noticeable spurs. However, when the input value is 16, the proposed MASH has approximately 10-dB higher noise than that of the HK-MASH at high frequencies. It is more clearly observed in Fig. 12(a) and (b), which show the

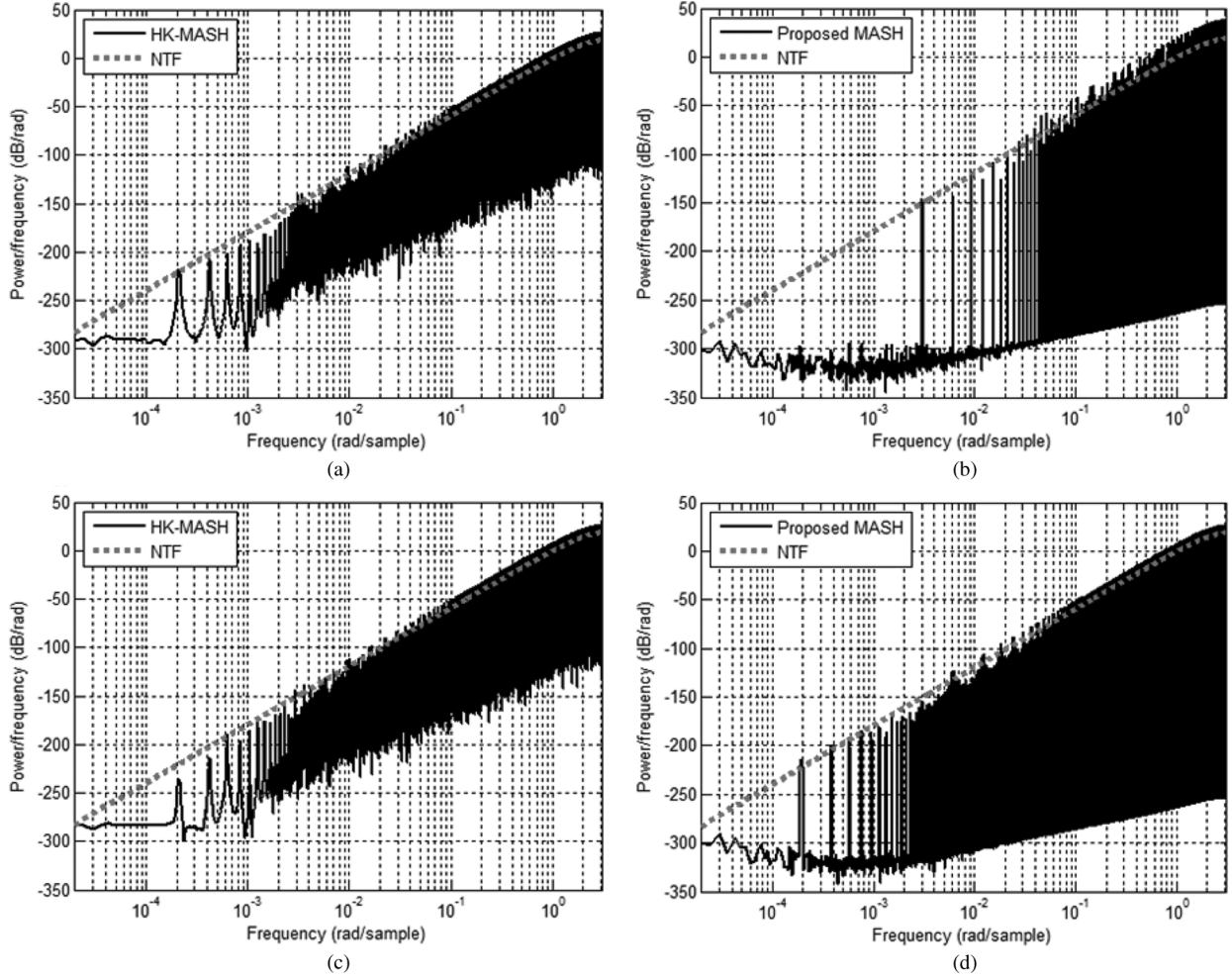


Fig. 10. Output spectra for input values of 16 and 17. (a) HK-MASH 1-1-1 with 5-bit EFMs for an input value of 16. (b) Proposed MASH 1-1-1 with 5-bit EFMs for an input value of 16. (c) HK-MASH 1-1-1 for 17. (d) Proposed MASH 1-1-1 for 17.

magnified spectra of Fig. 10. In the fractional- N frequency synthesizer, however, the DSM output signal is eventually applied to a low-pass filter that passes only the low-frequency components. Moreover, the passband cutoff frequency of the low-pass filter is normally much lower than the sampling frequency of the DSM, so the large noise spectrum at high frequencies can be removed by employing the low-pass filter. In addition, in order to reduce the noise at high frequencies, the proposed MASH 1-1-1 is restructured based on Corollary 1, as shown in Fig. 8. The sequence length of the proposed MASH with one 5-bit EFM at the first stage and 9-bit EFMs at the later stages is equal to 2^{19} for the input value 16. Moreover, the noise spectrum at high frequencies is reduced and similar to that of the HK-MASH, as depicted in Fig. 12(c).

The effect of the proposed MASH structure is investigated in a fractional- N frequency synthesizer as well. The frequency synthesizer is simulated by using CppSim [18]. Given a 20-MHz reference clock, the frequency synthesizer generates an output frequency of 1.855 GHz, which corresponds to 92.75 times the reference frequency. We assume that the reset delay of the phase and frequency detector is 3.9 ns, and the charge pump current is 24 μ A. Moreover, the voltage-controlled oscillator has a center frequency of 1.84 GHz and a phase noise of -130 dBc/Hz at

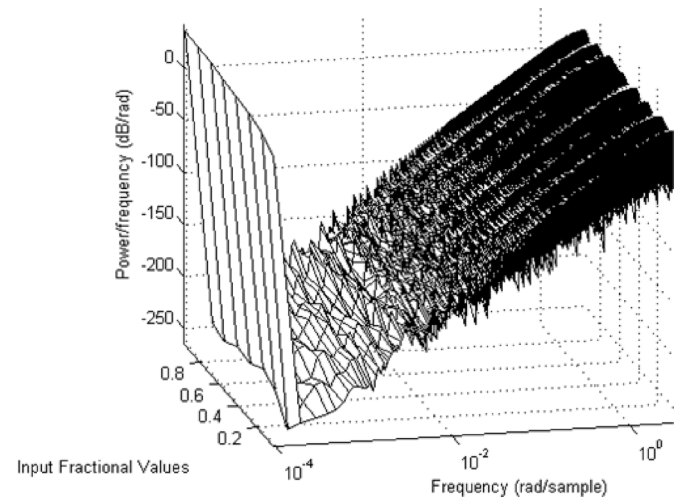


Fig. 11. Output spectra of the proposed MASH 1-1-1 with 5-bit EFMs for several input fractional values.

10 MHz. In addition, the input bitwidth of the DSM is set to 5 bits. The traditional MASH 1-1-1 with three 5-bit EFMs and the proposed MASH 1-1-1 with three 5-bit EFMs are simulated, and

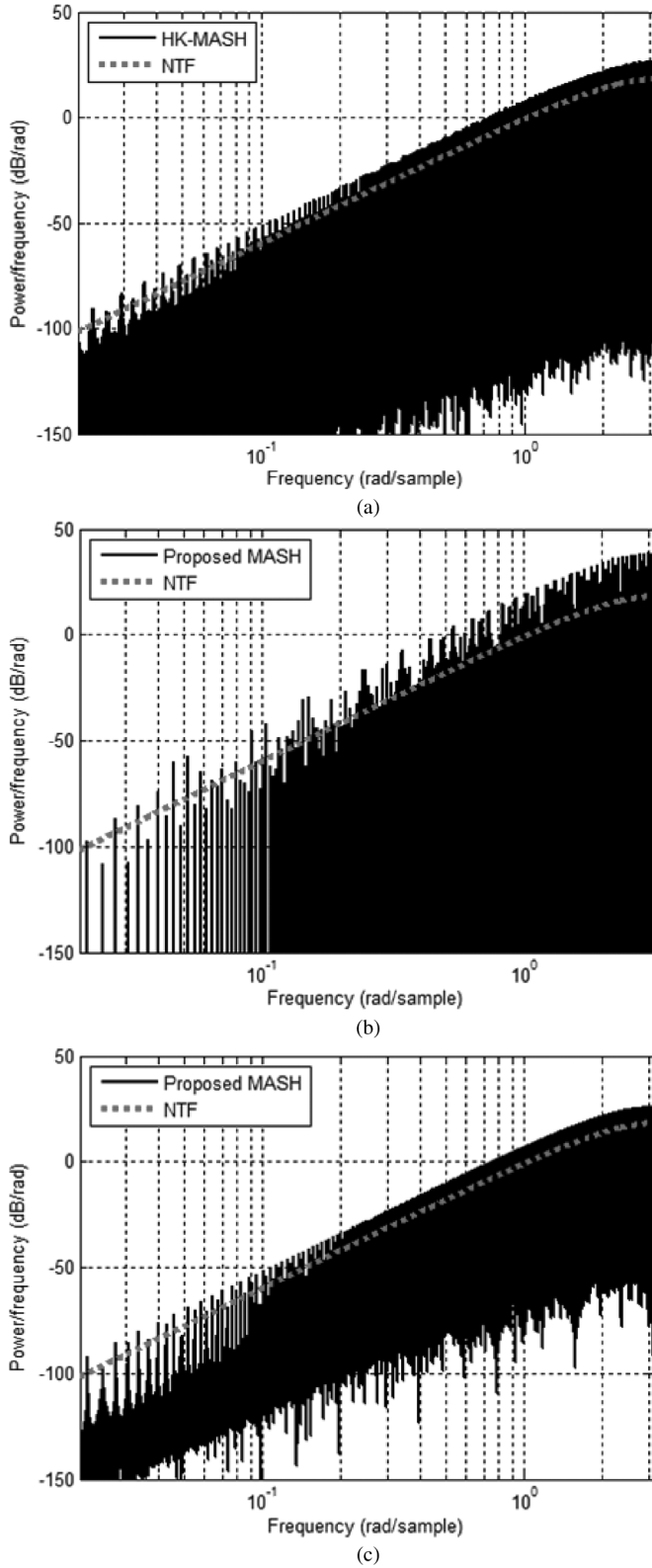


Fig. 12. Detailed output spectra for an input value of 16. (a) HK-MASH 1-1-1 with 5-bit EFMs. (b) Proposed MASH 1-1-1 with 5-bit EFMs. (c) Proposed MASH 1-1-1 with one 5-bit EFM at the first stage and 9-bit EFMs at the later stages.

the phase noise spectra resulting from the frequency synthesizer are shown in Fig. 13. All fractional spurs, except the reference spur at 20 MHz in Fig. 13(a), are removed by employing the

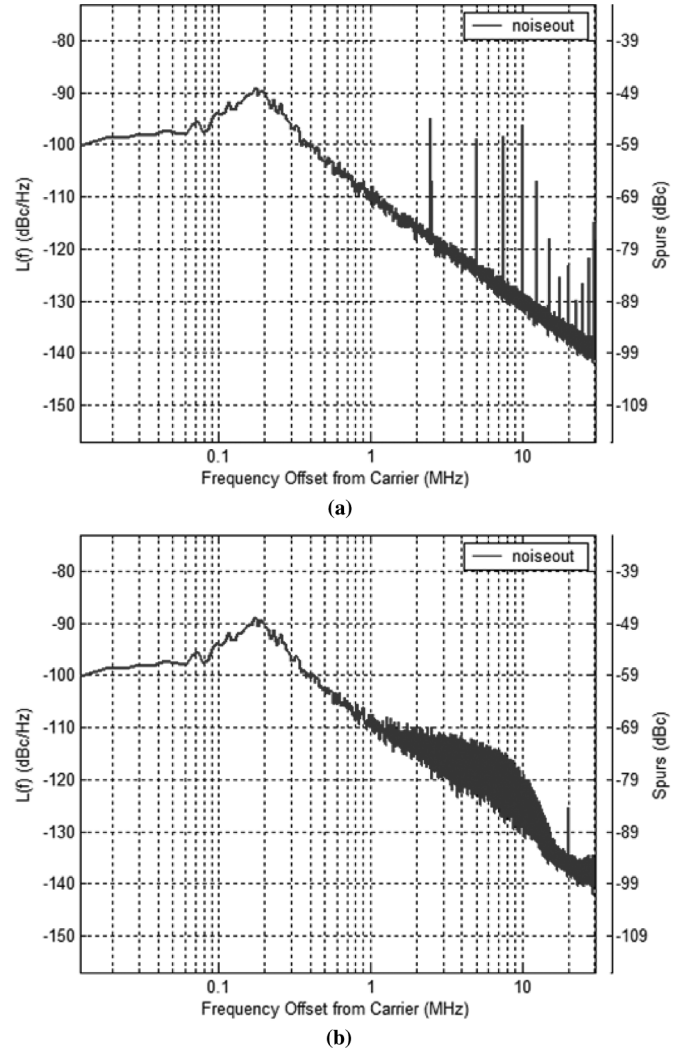


Fig. 13. Phase noise spectra of a fractional- N frequency synthesizer employing (a) the traditional MASH 1-1-1 with 5-bit EFMs and (b) the proposed MASH 1-1-1 with 5-bit EFMs.

proposed MASH DSM, as shown in Fig. 13(b). Note that the reference spur is caused by the reference clock and not by the DSM.

Table I compares several MASH 1-1-1 structures grounded on the deterministic approach. The minimum sequence length of the traditional MASH structure is very short, and the maximum sequence length is independent of the MASH order. The MASH structure proposed in [4] increases the sequence length to $2M$, and no conversion operations are necessary for the input values generated by a DSM controller. However, the sequence length is not long enough to distribute the quantization error power. In [6], the quantization interval P is set to the largest prime number less than M . The prime modulus is effective in increasing the minimum sequence length [6]. However, the sequence length rarely increases as the MASH order increases [6], and the input range is limited to $\{0, 1, \dots, (P - 1)\}$. Moreover, conversion operations are required since the desired fractional value should be scaled by P/M in generating the input value of the DSM. In [7], an efficient MASH structure called HK-MASH is presented based on a modified EFM in which the output is multiplied by

TABLE I
COMPARISON OF MASH 1-1-1 STRUCTURES

Architecture	Maximum Sequence Length	Minimum Sequence Length	Input Range
Traditional MASH	M^a	2	$0 \sim (M-1)$
Borkowski <i>et al.</i> [4]	$2M$	$2M$	$0 \sim (M-1)$
Hosseini <i>et al.</i> [6]	P^b	P	$0 \sim (P-1)$
HK-MASH [7]	$(M-a)^3$	$(M-a)^3$	$0 \sim (M-a-1)$
Proposed MASH	ML^{2c}	$2L^2$	$0 \sim (M-1)$

a $M = 2^b$ when the input fractional value is represented in b bits.

b P is a maximum prime number less than M .

c $L = 2^r$, where r is the word length of the second and third stages ($r \geq b$).

TABLE II
IMPLEMENTATION RESULTS OF THE MASH 1-1-1 STRUCTURES

Architecture	Word Length	Gate Counts	Power (mW)	Frequency (MHz)
HK-MASH [7]	5	377	1.296	217
	7	481	1.694	217
	9	835	2.683	217
	11	936	2.992	217
	13	946	3.215	217
	15	1491	4.747	217
Proposed MASH	5	355	1.252	217
	7	471	1.652	217
	9	596	2.091	217
	11	740	2.548	217
	13	857	2.965	217
	15	973	3.361	217

a and fed back to the input. The sequence length becomes long enough to achieve smooth spectra because the feedback in effect makes the largest prime number $(M - a)$ within the word length b . The average of the output sequence of the HK-MASH is linearly proportional to the input fractional value. However, the feedback loop makes the HK-MASH to have a bounded input range. Due to the same reason as [6], additional operations are indispensable in converting the input value. The conversion is a burden when achieving a low-complexity DSM [20],[21].

The sequence length of the proposed MASH structure is long enough to distribute the quantization error power. Moreover, the quantization intervals of the first stage and the other stages can be different, according to Corollary 1. When the moduli of the first stage and the other stages are M and L , respectively, the sequence length is ranged from $2L^2$ to $M \cdot L^2$ for the proposed MASH 1-1-1 structure. The proposed MASH structure does not require any conversion operations in interfacing a controller. As a result, the proposed MASH DSM is close to the ideal DSM.

To compare the hardware cost, third-order MASH DSMs are designed based on the proposed MASH and the HK-MASH structures and synthesized with 0.18- μ m CMOS technology. The hardware complexity is measured by counting the number of equivalent gates. In Table II, the gate counts and power consumptions of the proposed MASH and the HK-MASH are compared. All the MASHs are synthesized to work at 217 MHz. Compared to the HK-MASH, the gate count of the proposed MASH is 35% less at maximum, and the power consumption is 30% less at maximum. Theoretically, the proposed MASH structure always has the same hardware cost as the traditional MASH structure. An accumulator for each EFM in the proposed

MASH structure has three inputs, whereas it has only two inputs in the traditional MASH structure. As the additional input coming from the previous stage is a single bit, it can be connected to the carry input of the accumulator at the current stage. Therefore, it is possible to add three inputs without increasing the hardware cost.

V. CONCLUSION

A new MASH DSM has been presented to remove spurs in the output spectrum. In addition to the spur-free property, the output average of the proposed MASH is exactly equal to the input fractional value. The proposed MASH structure cascades several first-order EFMs like the traditional MASH structure but has an additional feedforward connection between adjacent stages. The additional feedforward connection has an effect of increasing the sequence length dramatically to $N_1 \cdot L^{l-1}$, where N_1 is the sequence length of the first stage, L is the modulus of the other stages, and l is the MASH order. We have proved this result mathematically and performed simulations to observe the noise-shaping effect resulting from the long sequence length. Moreover, the hardware complexity of the proposed MASH structure is not increased even compared to the traditional MASH structure.

APPENDIX

PROOFS OF PROPOSITIONS AND THEOREMS

To prove the sequence length of the proposed MASH structure, we first need to introduce Theorem 1 and Lemma 1 related to the EFM shown in Fig. 2(a). Theorem 1 has already been proved in [13], but its proof is described here again for the completeness of this paper. Next, we present Lemma 2 required to prove Theorem 2. Assume that M is 2^b and that α is the desired fractional value.

Proof of Theorem 1: From the definition of $e[n]$, we obtain

$$e[n] = (e[0] + n \cdot M \cdot \alpha) \bmod M \quad (17)$$

where $e[0]$ is an initial value of the register in the EFM. Since $e[n]$ is a periodic signal, if the period of $e[n]$ is N

$$e[N] = (e[0] + N \cdot M \cdot \alpha) \bmod M. \quad (18)$$

$$\text{As } e[0] = e[N]$$

$$(N \cdot M \cdot \alpha) \bmod M = 0. \quad (19)$$

Because α is a fractional number, $N \cdot \alpha = N(2^j \cdot B)/M$, and $N \cdot \alpha$ should be an integer. Therefore, the sequence length of the EFM is equal to $2^{b-j} = M/2^j$. $\square$

Lemma 1: The mean of the quantization error signal $e[n]$ of the first-order DSM is $\text{Mean}(\{e[n]\}) = (M \cdot \beta/2 + e[0]) \bmod M$, where $\beta = (1 - \alpha)$ if $\alpha \leq 0.5$; otherwise, $\beta = \alpha$ if $x[n]$ is an input signal $M \cdot \alpha$, where α is a desired fractional value.

Proof: Since $e[n]$ is a periodic signal with sequence length N , the average value of $e[n]$ can be established by using (17)

$$\frac{1}{N} \sum_{k=1}^N e[k] = \frac{1}{N} (M \cdot \alpha + 2M \cdot \alpha + \dots$$

$$+N \cdot M \cdot \alpha + N \cdot e[0]) \bmod M. \quad (20)$$

As $(N \cdot M \cdot \alpha) \bmod M = 0$ in (19), we can derive two solutions. First

$$\begin{aligned} \frac{1}{N} (M \cdot \alpha + 2M \cdot \alpha + \dots + (N-1)M \cdot \alpha) \bmod M \\ = \left(\frac{(N-1)M \cdot \alpha}{2} \right) \bmod M. \end{aligned} \quad (21)$$

Using Theorem 1, we obtain

$$\left(\frac{(N-1)M \cdot \alpha}{2} \right) \bmod M = \left(\frac{M \cdot B - M \cdot \alpha}{2} \right) \bmod M \quad (22)$$

where $B \bmod 2 = 1$. Note that $(M \cdot B)/2 \bmod M = M/2$. We can substitute (20) by (22) while $e[0] \bmod M = e[0]$, so

$$\frac{1}{N} \sum_{k=1}^N e[k] = \left(\frac{M(1-\alpha)}{2} + e[0] \right) \bmod M. \quad (23)$$

The other solution can be derived by differently expanding (20) as follows:

$$\begin{aligned} \frac{1}{N} (M \cdot \alpha + 2M \cdot \alpha + \dots + N \cdot M \cdot \alpha) \bmod M \\ = \frac{1}{N} \left(N \cdot M \cdot \alpha \cdot \frac{N}{2} + \frac{N}{2} M \cdot \alpha \right) \bmod M. \end{aligned} \quad (24)$$

As $(N \cdot M \cdot \alpha) \bmod M = 0$

$$\frac{1}{N} \sum_{k=1}^N e[k] = \left(\frac{M\alpha}{2} + e[0] \right) \bmod M. \quad (25)$$

If $e[0]$ is zero, the minimum mean value of the quantization error signal is $M/4$ when $\alpha = 0.5$. Thus, the first solution (23) is valid when α is equal to or less than 0.5. Otherwise, the second solution (25) is valid. $\square$

Lemma 2: Let signal $a[k]$ be periodic and the sequence length N of $a[k]$ be a multiple of two. The double summation of $a[k]$ for M times N results in a value that is a multiple of M , namely

$$\left(\sum_{n=1}^{M \cdot N} \sum_{k=1}^n a[k] \right) \bmod M = 0. \quad (26)$$

Proof: Let us suppose that

$$\sum_{n=1}^N \sum_{k=1}^n a[k] = K \quad (27)$$

$$\sum_{k=1}^N a[k] = P \quad (28)$$

where K and P are constant values. Rearranging (26) with (27) and (28) gives

$$\sum_{n=1}^{M \cdot N} \sum_{k=1}^n a[k] = \sum_{n=1}^N \sum_{k=1}^n a[k] + \sum_{n=N+1}^{2N} \sum_{k=1}^n a[k] + \dots$$

$$\begin{aligned} &+ \sum_{n=(M-1)N+1}^{M \cdot N} \sum_{k=1}^n a[k] \\ &= K + (K + N \cdot P) + \dots + (K + (M-1)N \cdot P) \\ &= M \cdot K + \frac{M(M-1)}{2} N \cdot P. \end{aligned} \quad (29)$$

As the last equation of (29) is an integer multiple of M , we conclude (26). $\square$

We first consider the proposed MASH 1-1-1 structure as a subset of the proposed MASH shown in Fig. 6 and then generalize the result later. The sequence length N_1 of the first stage EFM is the same as the period of the conventional EFM according to Theorem 1.

Proof of Proposition 1: The quantization error of the second stage $e_2[n]$ is

$$\begin{aligned} e_2[n] &= (e_1[n] + y_1[n] + e_2[n-1]) \bmod M \\ &= \left(e_2[0] + \sum_{k=1}^n (e_1[k] + y_1[k]) \right) \bmod M. \end{aligned} \quad (30)$$

As $e_2[n]$ is periodic, $e_2[0] = e_2[N_2]$, where N_2 is the sequence length of the second stage. We can rewrite (30) as

$$\left(\sum_{n=1}^{N_2} (e_1[n] + y_1[n]) \right) \bmod M = 0 \quad (31)$$

where $e_1[n]$ and $y_1[n]$ are the outputs of the first stage. N_2 is the integer proportion of N_1 , so N_2 is equal to $L \cdot N_1$, where L is an integer. $e_1[n]$ and $y_1[n]$ are periodic with sequence length N_1 . We obtain

$$\begin{aligned} &\left(\sum_{n=1}^{L \cdot N_1} (e_1[n] + y_1[n]) \right) \bmod M \\ &= L \cdot \left(\sum_{n=1}^{N_1} (e_1[n] + y_1[n]) \right) \bmod M = 0. \end{aligned} \quad (32)$$

According to Lemma 1 and (5), we can rewrite (32) as

$$L \cdot \left(\frac{M \cdot \beta}{2} \cdot N_1 + N_1 \cdot e_1[0] + N_1 \cdot \alpha \right) \bmod M = 0 \quad (33)$$

where β is defined in Lemma 1. Since N_1 is a multiple of two, the modulo operation of the first term is zero, and L is concluded as the power of two. If $e_1[0]$ is zero, the term related to $e_1[0]$ is removed. Otherwise, L is still concluded as the power of two. By Theorem 1, $(N_1 \cdot \alpha) \bmod M = B \bmod M = B$. As a result, L should be M to satisfy that the third term of (33) is zero, so the sequence length of the second stage is $N_1 \cdot M$.

As the quantization error of the third stage $e_3[n]$ is also periodic, the relation

$$\left(\sum_{n=1}^{N_3} (e_2[n] + y_2[n]) \right) \bmod M = 0 \quad (34)$$

should be satisfied, where N_3 is the sequence length of the third stage. N_3 is the proportion of N_2 , so N_3 is equal to $L \cdot N_2$, where L is an integer. We obtain

$$L \cdot \left(\sum_{n=1}^{N_2} (e_2[n] + y_2[n]) \right) \bmod M = 0. \quad (35)$$

Substituting $e_2[n]$ with (30) gives

$$L \cdot \left(\sum_{n=1}^{M \cdot N_1} \left(\sum_{k=1}^n (e_1[k] + y_1[k]) + e_2[0] \right) + \sum_{n=1}^{N_2} y_2[n] \right) \times \bmod M = 0. \quad (36)$$

Since $e_1[k]$ and $y_1[k]$ are periodic, the modulo operation of the first term in (36) is zero according to Lemma 2. Moreover, the modulo operation of $M \cdot N_1 \cdot e_2[0]$ is zero. From (4), moreover, the summation of $y_2[n]$ is proportional to the summation of $e_1[n]$ and $y_1[n]$. Finally, we reduce (36) into

$$L \cdot \left(\sum_{n=1}^{N_1} (e_1[n] + y_1[n]) \right) \bmod M = 0 \quad (37)$$

which is the same result as (32), and L should be M . Therefore, the sequence length of the third stage N_3 is equal to $N_1 \cdot M^2$. $\square$

For example, let us consider the MASH 1-1-1 structure with 5-bit word length. For an input value of 16, the sequence lengths of the first, second, and third stages are 2, 64, and 2048, respectively, according to Theorem 1 and Proposition 1.

Using Proposition 1, we can generalize the sequence length of the l th stage based on the mathematical induction.

Proof of Theorem 2: Assuming that the sequence length of the $(l-1)$ th stage is known, $N_{l-1} = N_1 \cdot M^{l-2}$. Since the quantization error of the l th stage $e_l[n]$ is periodic, we obtain the relation

$$\left(\sum_{n=1}^{N_l} (e_{l-1}[n] + y_{l-1}[n]) \right) \bmod M = 0 \quad (38)$$

where N_l is equal to $L \cdot N^{l-1}$. The quantization error and the output signal of the $(l-1)$ th stage $e_{l-1}[n]$ and $y_{l-1}[n]$ are periodic. We can rewrite (38) as

$$L \cdot \left(\sum_{n=1}^{N_{l-1}} (e_{l-1}[n] + y_{l-1}[n]) \right) \bmod M = 0. \quad (39)$$

The quantization error of the $(l-1)$ th-stage $e_{l-1}[n]$ is replaced with the summation of the inputs of the $(l-1)$ th stage. We obtain

$$L \cdot \left(\sum_{n=1}^{M \cdot N_{l-2}} \left(\sum_{k=1}^n (e_{l-2}[k] + y_{l-2}[k]) + e_{l-1}[0] \right) + \sum_{n=1}^{N_{l-1}} y_{l-1}[n] \right) \bmod M = 0. \quad (40)$$

Since $e_{l-2}[k]$ and $y_{l-2}[k]$ are periodic, the modulo operation of the first term in (40) is zero according to Lemma 2. Moreover, the modulo operation of $M \cdot N_{l-2} \cdot e_{l-1}[0]$ is zero. The summation of $y_{l-1}[n]$ is proportional to the summation of $e_{l-2}[n]$ and $y_{l-2}[n]$ according to (4). Finally, we obtain

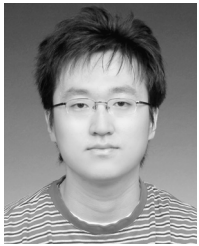
$$L \cdot \left(\sum_{n=1}^{N_{l-2}} (e_{l-2}[n] + y_{l-2}[n]) \right) \bmod M = 0. \quad (41)$$

When we repeat this procedure, we can conclude the same result as (32), so L is equal to M . As a result, the sequence length of the output signal in the proposed l th-order MASH structure is $N_1 \cdot M^{l-1}$. $\square$

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